Practice – 6

1. Objectives

A. Working with Dynamic Memory

2. Instructions

In this assignment, you will extend the provided Vector class to implement additional functionality:

1) int getSize() const. Returns current number of elements in the vector.
2) int getCapacity() const. Returns the capacity of the vector.
3) void clear(). Removes all elements from the vector. The method should deallocate the old memory, set size to 0 and allocate new memory with default capacity 10.
4) void push_front(int value). Insert value at the beginning of the vector (index 0). Shift all existing elements right to make space. Resize if capacity is exceeded (double the capacity)
5) void insert(int pos, int value). Insert value at position pos. Support negative indices: Convert -1 to size, -2 to size-1, etc. Shift elements right from position pos to end. Resize if capacity is exceeded (double the capacity).
6) Modify at(int index). Keep existing positive index behavior. Add negative index support: -1 = last element, -2 = second-to-last, etc.

Run main.cpp to test the code.